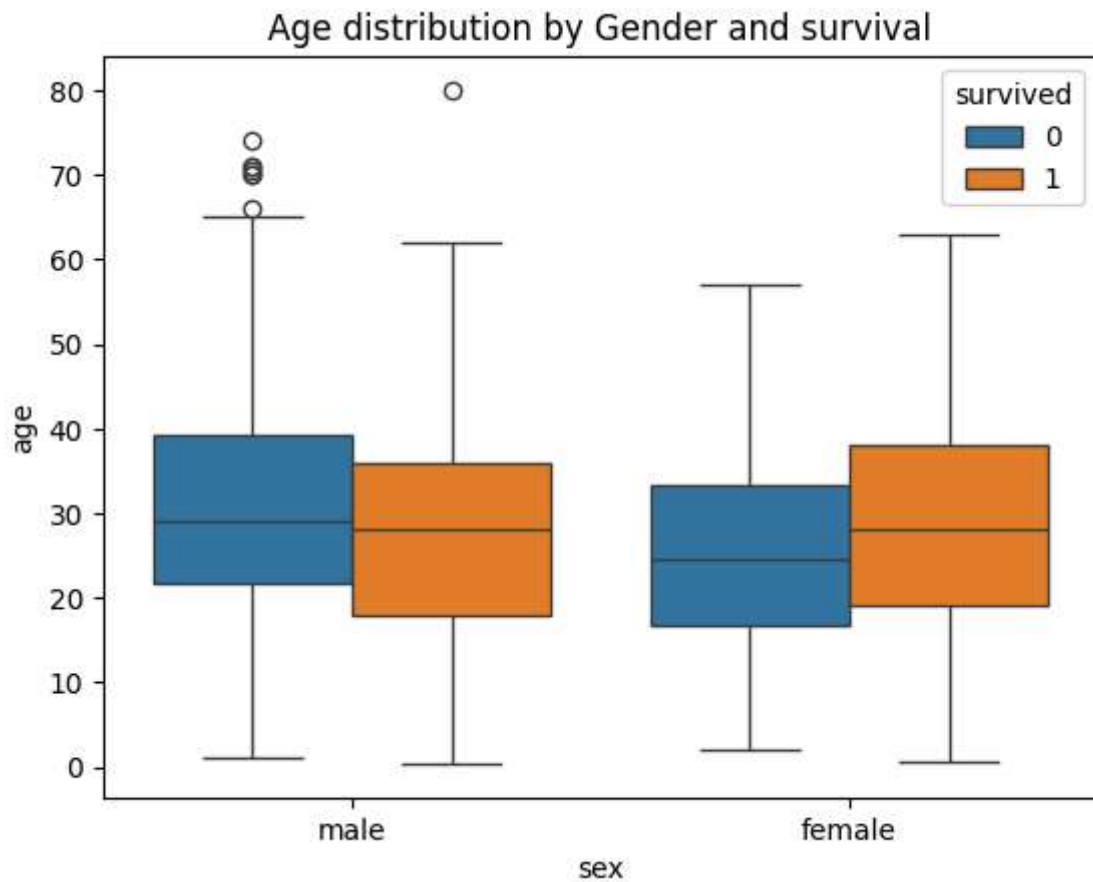


```
In [1]: import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [2]: df = sns.load_dataset('titanic')
```

```
In [5]: sns.boxplot(x='sex',y='age',hue='survived',data =df)
plt.title("Age distribution by Gender and survival")
plt.show()
#, the hue parameter is used to add a second categorical variable to your plot, eff
# the data by color to allow for deeper, nested comparisons
```



Observations from the Box Plot

1. Female Survival Rate is Higher

The orange box for females (`survived = 1`) is prominent, showing:

more females survived compared to males

This supports the historical:

"Women and children first"
policy during rescue.

2. Male Non-Survivors are More

For males:

- Blue box (`survived = 0`) is larger

Meaning:

more male passengers did not survive

3. Median Age of Survivors and Non-Survivors

The horizontal line inside each box represents:

median age

Observations:

- Median age of female survivors is around late 20s
- Median age of male survivors is also around late 20s

So:

many survivors were young adults

4. Male Age Distribution is More Spread

Male boxes and whiskers are wider.

Meaning:

greater variation in male passenger ages

Male passengers ranged from:

- children to
 - elderly people
-

5. Presence of Outliers

Dots above whiskers indicate:

outliers

Example:

- Male passengers around age 70–80

These are unusually high age values compared to majority.

6. Female Survivors Have Wider IQR

Female survivor box appears slightly larger.

Meaning:

female survivor ages are more spread

7. Children Also Survived

Lower whisker values near:

0–10 years

indicate:

children were present among survivors

8. Most Passenger Ages are Between 20–40

Most boxes are concentrated around:

20 to 40 years

Meaning:

majority passengers were young or middle-aged adults

Final Inference (Best Exam Answer)

"From the box plot, we observe that female passengers had higher survival rates than males. Male passengers who did not survive were more numerous. The median age of passengers is around 25–30 years. Male passengers show greater age variability, and some elderly passengers appear as outliers. Most passengers belonged to the age group of 20–40 years."

Viva Questions External May Ask

Q. What does the line inside box represent?

Median.

Q. What do circles/dots represent?

Outliers.

Q. Which gender had higher survival?

Females.

Q. Which gender shows more variability?

Males.

Q. What does bigger box indicate?

Larger IQR / higher spread.

DSBDA Practical 9 — Data Visualization II

Box Plot of Age Distribution by Gender and Survival

Dataset Used:

Titanic Dataset

Using:

Seaborn inbuilt Titanic dataset

Aim

To plot a box plot showing:

- Distribution of age
- With respect to gender
- Along with survival information

Columns used:

- sex
 - age
 - survived
-

What is a Box Plot?

Box plot is a graphical representation used to show:

- Distribution of data
 - Median
 - Quartiles
 - Outliers
 - Spread of data
-

Why Box Plot is Useful?

It helps identify:

- Skewness
 - Outliers
 - Data spread
 - Comparison between categories
-

Step 1: Import Libraries

Code

```
import seaborn as sns
import matplotlib.pyplot as plt
```

Explanation

Library	Purpose
seaborn	Statistical visualization
matplotlib	Plot display

Step 2: Load Titanic Dataset

Code

```
df = sns.load_dataset('titanic')
```

Display Dataset

Code

```
df.head()
```

Step 3: Plot Box Plot

Code

```
sns.boxplot(x='sex', y='age', hue='survived', data=df)

plt.title("Age Distribution by Gender and Survival")

plt.show()
```

Explanation of Each Command

`sns.boxplot()`

Creates box plot.

`x= 'sex'`

Separates plots by:

- male
- female

on X-axis.

`y= 'age'`

Displays age distribution on Y-axis.

`hue= 'survived'`

Further divides data based on:

- Survived
- Not survived

Usually:

0 = Not Survived

1 = Survived

```
data=df
```

Specifies dataset.

```
plt.title()
```

Adds graph title.

```
plt.show()
```

Displays graph.

Understanding Box Plot Components

1. Median Line

Middle horizontal line inside box.

Represents:

median age

2. Box

Represents:

Interquartile Range (IQR)

From:

- Q1 (25%) to
 - Q3 (75%)
-

3. Whiskers

Show:

normal range of data

4. Dots Outside Whiskers

Represent:

outliers

Important Theory

Quartiles

Quartile	Meaning
Q1	25%
Q2	Median
Q3	75%

IQR Formula

$$\text{IQR} = \text{Q3} - \text{Q1}$$

Observations / Inference

You should write observations like this in exam.

Observation 1

Female survival rate appears higher than male survival rate.

Reason:

- More females survived.
-

Observation 2

Median age differs between:

- males
 - females
 - survivors
 - non-survivors
-

Observation 3

Some outliers are present in age distribution.

These appear as:

dots outside whiskers

Observation 4

Children had relatively better survival chances.

(You may observe lower age survivors.)

Observation 5

Male age distribution is more spread compared to females.

Meaning:

higher variability

Important Viva Explanation

You can say:

"The box plot shows distribution of age with respect to gender and survival status. Median age, spread, and outliers can be observed clearly. Female passengers show higher survival rates, while male passengers have wider age distribution."

Why Use Hue?

Hue adds:

another categorical dimension

Here:

survived
creates separate color groups.

Important Viva Questions

Basic Questions

What is box plot?

Graphical representation of data distribution.

What does box represent?

Interquartile range.

What does line inside box represent?

Median.

What are whiskers?

Range of normal data.

What are dots outside whiskers?

Outliers.

Intermediate Questions

Why box plot preferred?

Easy comparison of distributions.

What is IQR?

Difference between Q3 and Q1.

Why hue used?

To add additional categorical grouping.

What does wider box indicate?

Higher variability.

Advanced Questions

Why female survival rate higher?

“Women and children first” rescue policy.

How to identify skewness in boxplot?

Median position and whisker lengths.

Why outliers important?

They indicate unusual observations.

External May Ask Modifications

Plot Without Survival

```
sns.boxplot(x='sex', y='age', data=df)

plt.show()
```

Plot Fare Distribution

```
sns.boxplot(x='sex', y='fare', data=df)

plt.show()
```

Horizontal Boxplot

```
sns.boxplot(x='age', y='sex', data=df)

plt.show()
```

Add Different Palette

```
sns.boxplot(x='sex', y='age', hue='survived',
            data=df, palette='Set2')

plt.show()
```

Difference Between Histogram and Boxplot

Histogram	Boxplot
Shows frequency	Shows statistical summary
Good for distribution	Good for comparison

Common Exam Errors

Error	Reason
Wrong column name	Case-sensitive
Missing hue variable	Wrong parameter
Null ages	Some rows have missing age

2-Minute Viva Explanation

"In this practical, we used Seaborn's Titanic dataset and plotted a box plot to visualize age distribution with respect to gender and survival status. The x-axis represents gender, the y-axis represents age, and hue represents survival information. The box plot displays median, quartiles, spread, and outliers. From the graph, we observed that female passengers had higher survival rates and male passengers showed wider age variation. Box plots are useful for comparing distributions across multiple categories."